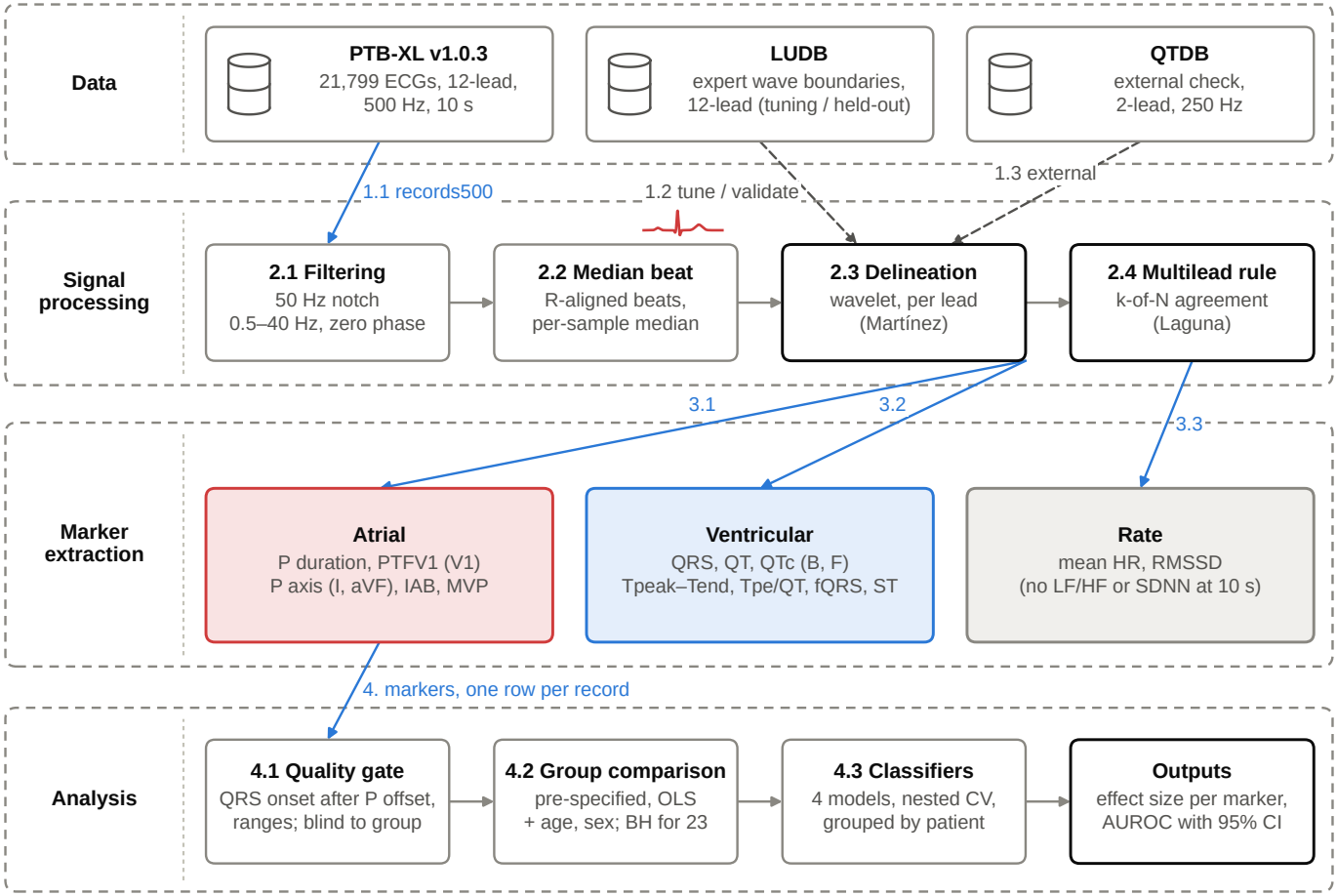
Data
PTB-XL v1.0.3
21,799 ECGs, 12-lead,
500 Hz, 10 s
LUDB
expert wave boundaries,
12-lead (tuning / held-out)
QTDB
external check,
2-lead, 250 Hz
1.1 records500
1.2 tune / validate
1.3 external
Signal processing
2.1 Filtering
50 Hz notch
0.5–40 Hz, zero phase
2.2 Median beat
R-aligned beats,
per-sample median
2.3 Delineation
wavelet, per lead
(Martínez)
2.4 Multilead rule
k-of-N agreement
(Laguna)
3.1
3.2
3.3
Marker extraction
Atrial
P duration, PTFV1 (V1)
P axis (I, aVF), IAB, MVP
Ventricular
QRS, QT, QTc (B, F)
Tpeak–Tend, Tpe/QT, fQRS, ST
Rate
mean HR, RMSSD
(no LF/HF or SDNN at 10 s)
4. markers, one row per record
Analysis
4.1 Quality gate
QRS onset after P offset,
ranges; blind to group
4.2 Group comparison
pre-specified, OLS
+ age, sex; BH for 23
4.3 Classifiers
4 models, nested CV,
grouped by patient
Outputs
effect size per marker,
AUROC with 95% CI